In [2]:
```
import pandas as pd
from sqlalchemy import create_engine

# ── 1. CONNECT ──────────────────────────────────────────────────
engine = create_engine('mysql+pymysql://root:0000@localhost/saudi_maroof')

# ── 2. LOAD CSV ─────────────────────────────────────────────────
df = pd.read_csv('saudi_ecommerce_maroof.csv', encoding='utf-8-sig')
print("Raw shape:", df.shape)
print("Columns:", df.columns.tolist())

# ── 3. CLEAN COLUMN NAMES ───────────────────────────────────────
df.columns = df.columns.str.strip().str.replace(' ', '_')

# ── 4. ENGINEER FEATURES ────────────────────────────────────────
df['has_website']   = df['Website'].notna().astype(int)
df['has_instagram'] = df['Instagram'].notna().astype(int)
df['has_twitter']   = df['Twitter'].notna().astype(int)
df['has_email']     = df['Email'].notna().astype(int)
df['has_phone']     = df['Phone_Number'].notna().astype(int)
df['digital_score'] = df[['has_website','has_instagram',
                          'has_twitter','has_email','has_phone']].sum(axis=1)
df['Rating']     = pd.to_numeric(df['Rating'],      errors='coerce')
df['Num_Ratings']= pd.to_numeric(df['Num_Ratings'], errors='coerce').fillna(0).a

# ── 5. LOAD dim_categories FIRST ────────────────────────────────
# Get every unique category and assign an ID
categories = df['Category'].dropna().unique()
dim_categories = pd.DataFrame({
    'category_id':   range(1, len(categories) + 1),
    'category_name': categories
})
dim_categories.to_sql('dim_categories', con=engine,
                      if_exists='append',   # append so AUTO_INCREMENT IDs stay
                      index=False)
print(f"✅ dim_categories loaded: {len(dim_categories)} categories")

# ── 6. MAP category_id BACK TO MAIN DF ──────────────────────────
df = df.merge(dim_categories, left_on='Category', right_on='category_name', how=

# ── 7. BUILD fact_stores ────────────────────────────────────────
# Match EXACTLY the columns you created in your schema
fact_stores = df[[
    'Name_ENG',
    'Name_AR',
    'CR',
    'category_id',
    'has_website',
    'has_instagram',
    'has_twitter',
    'has_email',
    'has_phone',
    'Activity',
    'Rating',
    'Num_Ratings',
    'digital_score'
]].rename(columns={
    'Name_ENG':  'name_eng',
```

```
    'Name_AR':   'name_ar',
    'CR':        'cr_number',
    'Activity':  'activity_status',
    'Rating':    'rating',
    'Num_Ratings':'num_ratings'
})

fact_stores.to_sql('fact_stores', con=engine,
                   if_exists='append',   # append so store_id AUTO_INCREMENT wo
                   index=False,
                   chunksize=500)
print(f"✅ fact_stores loaded: {len(fact_stores)} stores")

# ── 8. VERIFY ──────────────────────────────────────────────
check = pd.read_sql("SELECT COUNT(*) AS total FROM fact_stores", engine)
print("Rows in MySQL:", check['total'][0])
```

```
Raw shape: (5401, 12)
Columns: ['Name_ENG', 'Name_AR', 'CR', 'Category', 'Website', 'Instagram', 'Twitt
er', 'Email', 'Phone Number', 'Activity', 'Rating', 'Num_Ratings']
✅ dim_categories loaded: 15 categories
✅ fact_stores loaded: 5401 stores
Rows in MySQL: 5401
```

In [3]:
```
fact_stores.head()
```

Out[3]:

	name_eng	name_ar	cr_number	category_id	has_website	has_instagram	has_twi
0	SpeshBox	سبيش بوكس	NaN	1	1	1	
1	NaN	متجر بيع عملات رقميه بيتكوين	NaN	2	1	0	
2	NaN	شحن شدات هاكات هكر ببجي	NaN	2	1	0	
3	NaN	تنسيق سهرات من المملكه العربيه	NaN	3	1	0	
4	Business Hub	شركة ملتقى الأعمال الإبداعية للإستثمار	4.650545e+09	4	1	0	

In [4]:
```
fact_stores.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 5401 entries, 0 to 5400
Data columns (total 13 columns):
 #   Column           Non-Null Count  Dtype
---  ------           --------------  -----
 0   name_eng         5010 non-null   str
 1   name_ar          5401 non-null   str
 2   cr_number        1910 non-null   float64
 3   category_id      5401 non-null   int64
 4   has_website      5401 non-null   int64
 5   has_instagram    5401 non-null   int64
 6   has_twitter      5401 non-null   int64
 7   has_email        5401 non-null   int64
 8   has_phone        5401 non-null   int64
 9   activity_status  5401 non-null   str
 10  rating           5401 non-null   float64
 11  num_ratings      5401 non-null   int64
 12  digital_score    5401 non-null   int64
dtypes: float64(2), int64(8), str(3)
memory usage: 548.7 KB
```

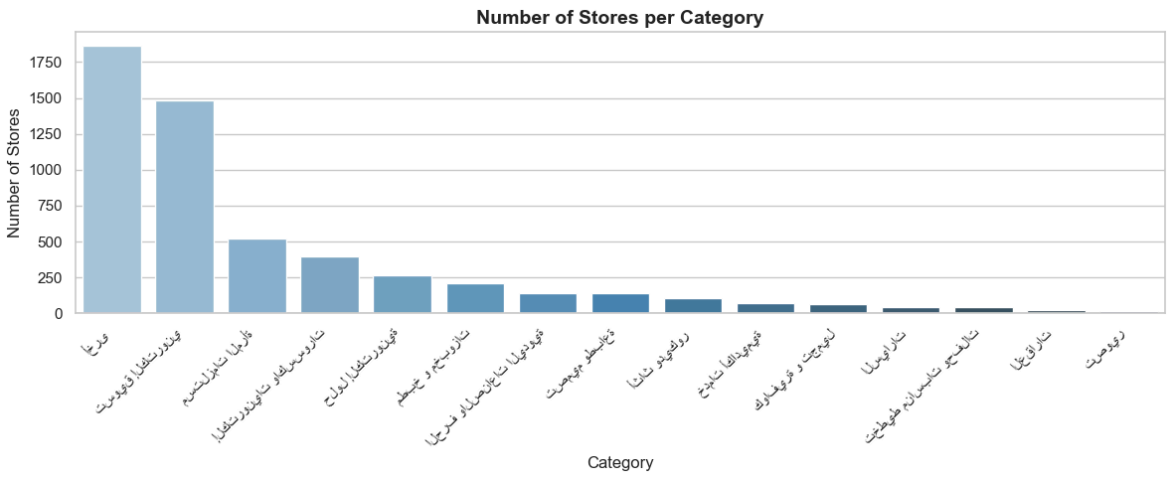
In [9]:
```
fact_stores.shape
```

Out[9]:
```
(5401, 13)
```

In [11]:
```
import matplotlib.pyplot as plt
import seaborn as sns
```

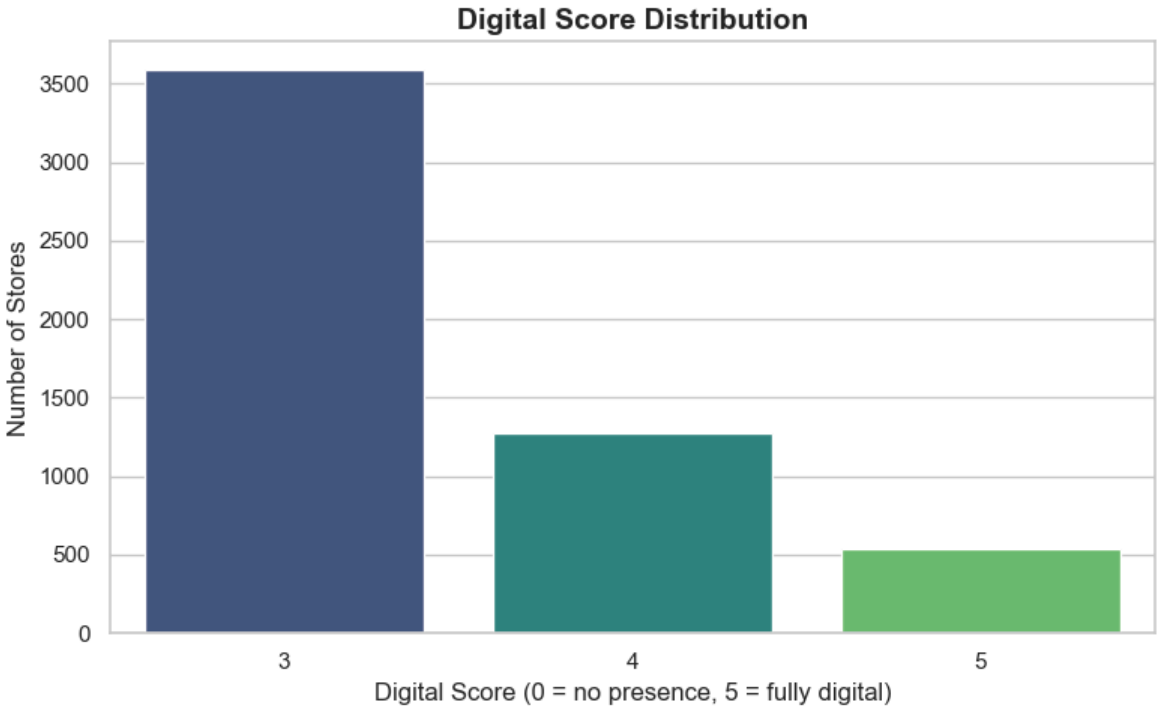
In [13]:
```
# ═══════════════════════════════════════════════════════════════════
# ── 9. VISUALIZATIONS ──────────────────────────────────────────
# ═══════════════════════════════════════════════════════════════════

sns.set_theme(style='whitegrid')  # clean background for all charts

# ── VIZ 1: Stores per Category ─────────────────────────────────
plt.figure(figsize=(12, 5))
category_counts = df['Category'].value_counts()
sns.barplot(x=category_counts.index, y=category_counts.values, palette='Blues_d'
plt.title('Number of Stores per Category', fontsize=14, fontweight='bold')
plt.xlabel('Category')
plt.ylabel('Number of Stores')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

```
C:\Users\mehmo\AppData\Local\Temp\ipykernel_14008\2073399238.py:10: FutureWarnin
g:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.barplot(x=category_counts.index, y=category_counts.values, palette='Blues_
d')
```



In [14]:
```
# ── VIZ 2: Digital Score Distribution ──────────────────────────
plt.figure(figsize=(8, 5))
sns.countplot(x='digital_score', data=df, palette='viridis')
plt.title('Digital Score Distribution', fontsize=14, fontweight='bold')
plt.xlabel('Digital Score (0 = no presence, 5 = fully digital)')
plt.ylabel('Number of Stores')
plt.tight_layout()
plt.show()
```

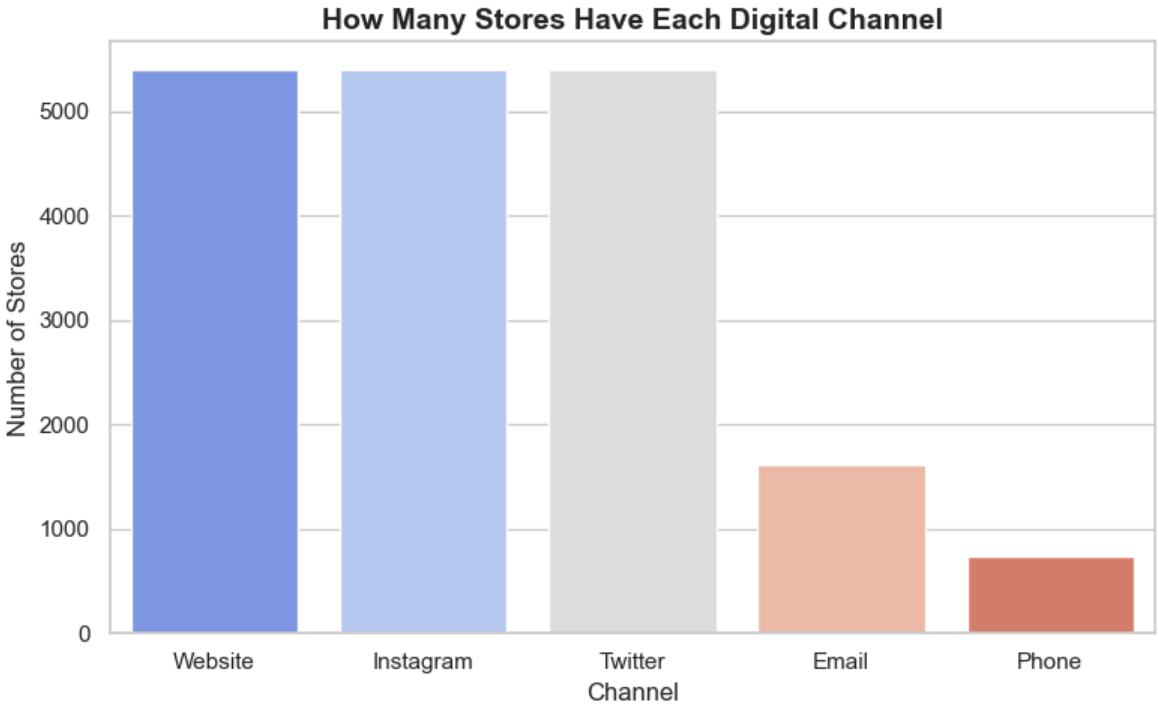
```
C:\Users\mehmo\AppData\Local\Temp\ipykernel_14008\2935660042.py:3: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.countplot(x='digital_score', data=df, palette='viridis')
```



In [15]:
```
# ── VIZ 3: Digital Channel Adoption ────────────────────────────
plt.figure(figsize=(8, 5))
channels = ['has_website', 'has_instagram', 'has_twitter', 'has_email', 'has_pho
channel_sums = df[channels].sum().sort_values(ascending=False)
channel_labels = ['Website', 'Instagram', 'Twitter', 'Email', 'Phone']
sns.barplot(x=channel_labels, y=channel_sums.values, palette='coolwarm')
```

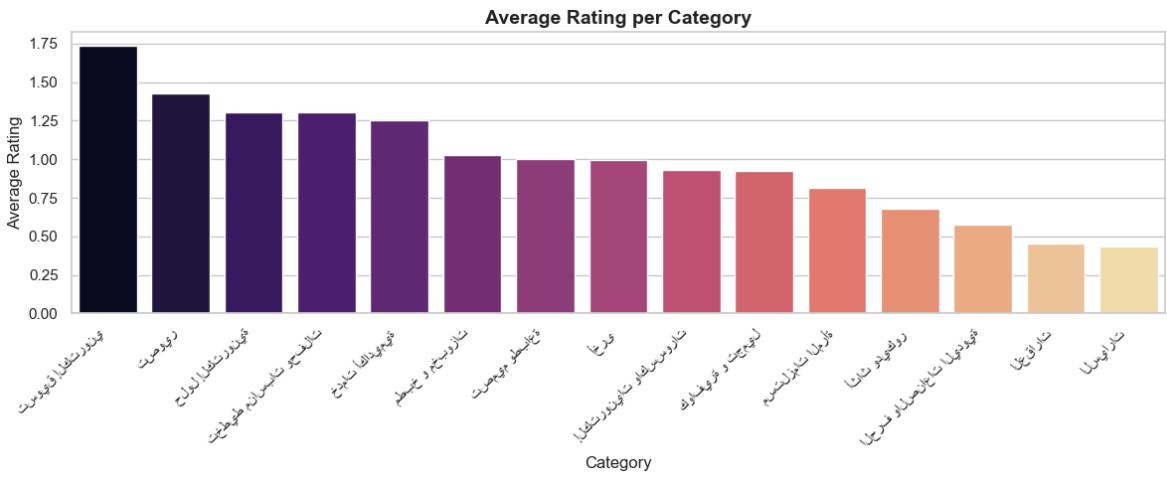
```
plt.title('How Many Stores Have Each Digital Channel', fontsize=14, fontweight='
plt.xlabel('Channel')
plt.ylabel('Number of Stores')
plt.tight_layout()
plt.show()
```

```
C:\Users\mehmo\AppData\Local\Temp\ipykernel_14008\2470217058.py:6: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.barplot(x=channel_labels, y=channel_sums.values, palette='coolwarm')
```

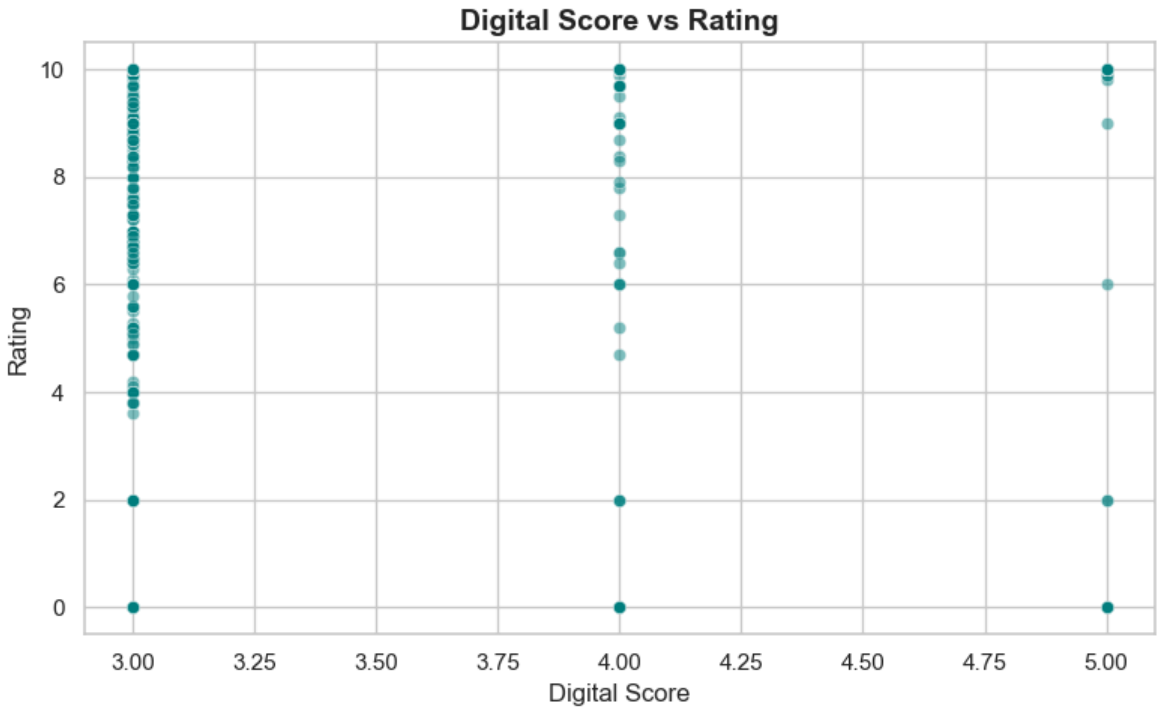


In [17]:
```
# ── VIZ 5: Average Rating per Category ──────────────────────────
plt.figure(figsize=(12, 5))
avg_rating = df.groupby('Category')['Rating'].mean().sort_values(ascending=False
sns.barplot(x=avg_rating.index, y=avg_rating.values, palette='magma')
plt.title('Average Rating per Category', fontsize=14, fontweight='bold')
plt.xlabel('Category')
plt.ylabel('Average Rating')
plt.xticks(rotation=45, ha='right')
plt.tight_layout()
plt.show()
```

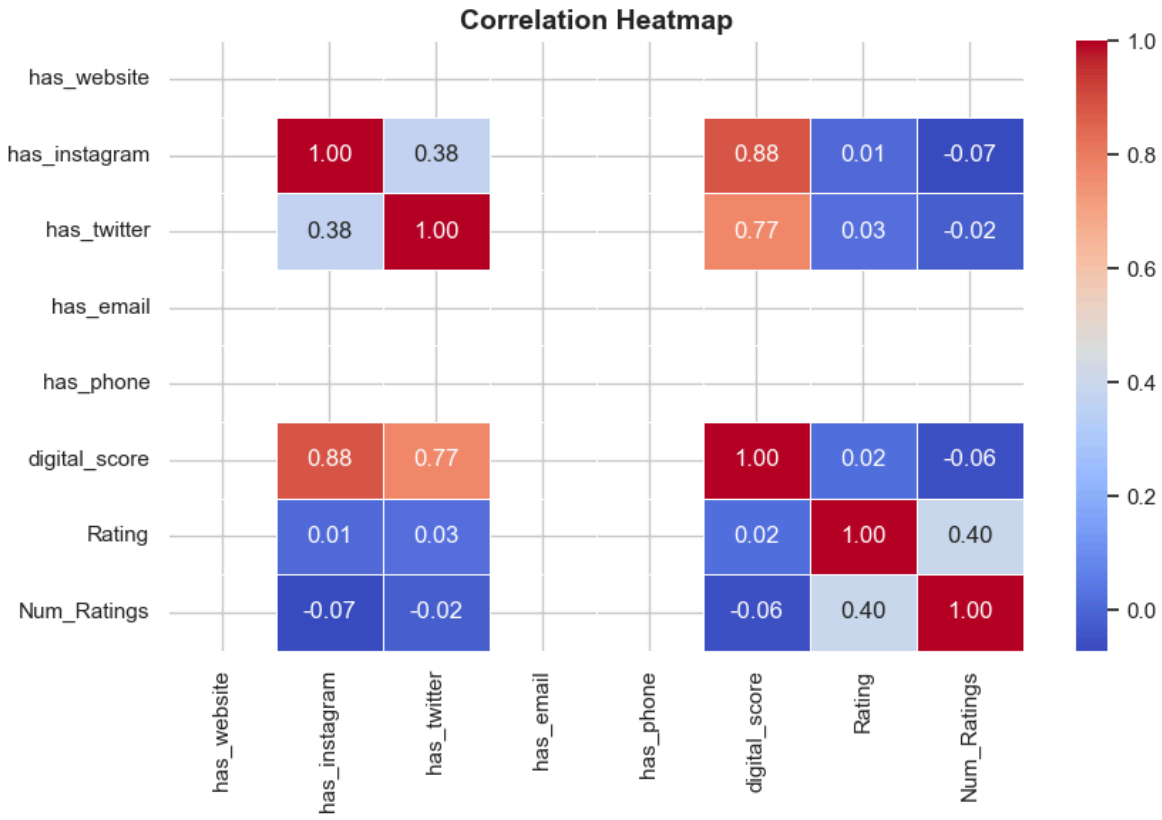
```
C:\Users\mehmo\AppData\Local\Temp\ipykernel_14008\2743357337.py:4: FutureWarning:

Passing `palette` without assigning `hue` is deprecated and will be removed in v
0.14.0. Assign the `x` variable to `hue` and set `legend=False` for the same effe
ct.

  sns.barplot(x=avg_rating.index, y=avg_rating.values, palette='magma')
```



In [18]:
```
# ── VIZ 6: Digital Score vs Rating (Scatter) ──────────────────
plt.figure(figsize=(8, 5))
sns.scatterplot(x='digital_score', y='Rating', data=df, alpha=0.5, color='teal')
plt.title('Digital Score vs Rating', fontsize=14, fontweight='bold')
plt.xlabel('Digital Score')
plt.ylabel('Rating')
plt.tight_layout()
plt.show()
```



In [19]:
```
# ── VIZ 7: Correlation Heatmap ────────────────────────────────
plt.figure(figsize=(9, 6))
numeric_cols = ['has_website', 'has_instagram', 'has_twitter',
                'has_email', 'has_phone', 'digital_score', 'Rating', 'Num_Rating
corr = df[numeric_cols].corr()
sns.heatmap(corr, annot=True, fmt='.2f', cmap='coolwarm', linewidths=0.5)
plt.title('Correlation Heatmap', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()
```


Correlation Heatmap
has_website
has_instagram
has_twitter
has_email
has_phone
digital_score
Rating
Num_Ratings
1.00 0.38 0.88 0.01 -0.07
0.38 1.00 0.77 0.03 -0.02
0.88 0.77 1.00 0.02 -0.06
0.01 0.03 0.02 1.00 0.40
-0.07 -0.02 -0.06 0.40 1.00
1.0
0.8
0.6
0.4
0.2
0.0